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Profile Summary

AI/ML Engineer and full-stack developer with 10+ years of self-taught programming experience (since 2014). Builds production systems spanning geospatial AI, hybrid RAG pipelines, generative models, and edge deployment — delivered for real clients including government institutions. Currently working as an AI Specialist at CubeTen and an LLM Evaluator at Outlier.ai, with a growing research interest in AI alignment, specifically mechanistic interpretability and mesa-optimization. Specializes in end-to-end development: data pipelines, model training, API design, and deployment.

Experience

CubeTen, Manipur

June 2025 – Present

AI Specialist

- Built a production hybrid document retrieval system for the **Manipur e-Gazette** — combining BM25 keyword search with pgvector semantic search over a 14GB+ corpus of government records, with source-cited answers and a custom OCR pipeline (Dolphin VLM) for degraded historical documents.
- Developed an AI-powered chatbot assistant for **Manipur eSeaba**, an official Government of Manipur citizen services portal, integrating RAG-based retrieval over government documentation.
- Provided IT support for the **12th Manipur Legislative Assembly Session** under NEVA and served as ticketing partner for **Sangai Festival 2025**.

Outlier.ai, Remote

October 2024 – Present

LLM Evaluator

- Evaluating large language model responses across RLHF alignment, truthfulness, safety, reasoning quality, and instruction following for frontier model training pipelines.
- Providing high-quality annotations supporting fine-tuning, safety training, and performance benchmarking.

Nibiaa Devices, Manipur

August 2024 – October 2024

AI Engineer

- Developed and optimized **real-time computer vision models** for EV bus passenger monitoring on NVIDIA Jetson Nano — full ML workflow from data preprocessing and model training to on-device deployment.
- Modified the Meshtastic open-source codebase to build a **P2P mesh networking Android app**, shipped to the Google Play Store in under two weeks with no prior Android experience.
- Contributed to two production applications launched on the **Google Play Store**.

NIT Manipur

May 2024 – July 2024

Summer Research Intern

- Supported development of an **English–Manipuri translation dataset** for low-resource machine translation research through proofreading and quality validation.
- Applied NLP preprocessing strategies to improve dataset quality and consistency.

Freelance

Dec 2020 – July 2024

Developer & ML Engineer

Manipur / Remote

- Built and deployed full-stack web applications for clients using **MERN, Django, and FastAPI** — including authentication systems, dashboards, and modular backend APIs.
- Developed ML features including classification models, NLP pipelines, and computer vision workflows for client projects.
- Mentored students and professionals in ML development — several mentees received outstanding grades on supervised projects.
- Maintained open-source projects selected for **SWOC 2025** and **IWOC 2025**; contributed to Hacktoberfest 2022 & 2024.

Projects

Manipur e-Gazette RAG System | Python, FastAPI, pgvector, BM25, Dolphin OCR, PostgreSQL

- Production hybrid retrieval QA system over a **14GB+ corpus** of government gazette records for a confidential client — combines BM25 lexical search with pgvector semantic search via reciprocal rank fusion.
- Custom OCR pipeline using **Dolphin VLM** to extract text from physically degraded historical documents where standard OCR fails.
- Returns AI-generated answers with **source citations** to specific gazette sections for fact verification.

Manipur Jal Sampada | Python, Sentinel-2, FLAIR, Rasterio, GIS, Google Colab

- Geospatial water encroachment detection system for a government client — learned GIS from scratch for this project.
- Initial NDWI/NDBI/NDVI threshold approach proved error-prone on real terrain; switched to **FLAIR ResNet34-UNet** on Sentinel-2 multispectral imagery, significantly improving classification accuracy.
- Generates semantic event masks (urban expansion, flooding, wetland conversion) exported as **GIS-ready layers** for further analysis.

Face2Comic – Pix2Pix cGAN | Python, PyTorch, U-Net, PatchGAN

- Conditional GAN translating real portrait photographs into comic-style illustrations — MCA final semester project. Model published on **HuggingFace**.
- **U-Net** generator with skip connections preserves spatial detail; **PatchGAN** discriminator enforces high-frequency texture realism over full-image classification.

Image Captioning Model | Python, TensorFlow, CNN, Transformer, Flickr8k

- Multimodal vision-language model combining **CNN encoder + Transformer decoder** with cross-modal attention for natural language caption generation — MCA minor project.
- Trained on Flickr8k under constrained compute (Google Colab); cross-modal attention learns which image regions are salient for each generated word.

EV Bus Monitoring System | Python, OpenCV, NVIDIA Jetson Nano

- Real-time people-counting and monitoring system optimized for **edge deployment** on NVIDIA Jetson Nano.
- Implemented efficient preprocessing, inference optimization, and hardware-level tuning for stable production performance.

Technical Skills

- **Programming:** Python, JavaScript, TypeScript
- **AI/ML:** PyTorch, TensorFlow, Scikit-learn, XGBoost, GANs, RAG, LangChain, Embeddings, NLP, Computer Vision, Geospatial AI, LLM Evaluation, RLHF, Mechanistic Interpretability (learning)
- **Backend & APIs:** FastAPI, Django, Flask, Node.js, Express.js, REST APIs, Microservices
- **Data & Pipelines:** Pandas, NumPy, pgvector, Vector Databases, PDF/Text Extraction, BM25, Rasterio
- **Databases:** PostgreSQL, MongoDB, MySQL, SQLite
- **Frontend:** React, Next.js, Vue.js, TailwindCSS
- **Cloud & DevOps:** AWS (EC2, S3), GCP, Docker, GitHub Actions
- **Tools:** Git, Google Colab, Streamlit, Gradio, QGIS

Education

NIELIT, Imphal Master of Computer Application (Grade: 71.00%)	June 2021 - July 2024 Imphal, Manipur
NIELIT, Imphal Bachelor of Computer Application (Grade: 64.83%)	June 2018 - June 2021 Imphal, Manipur

Achievements

- **Guest Speaker** – NIELIT Imphal Cybrella 2.0 (2025) on “Open Source Contribution” as Subject Matter Expert, invited to bring awareness of open source opportunities to students and professionals.
- **Open Source Maintainer** – Project maintainer for Social Winter of Code (SWOC) 2025 and Innogeeks Winter of Code (IWOC) 2025; reviewed contributor PRs and managed issue tracking across both programs.
- **Hacktoberfest Contributor** – 2022 and 2024.
- **AI/VR/AR Hackathon, MIMS MU (2025)** – Trained a ResNet50 race classification model on A100 GPU using the FairFace dataset; published trained model on HuggingFace.

Certifications

- **Machine Learning Specialization** – DeepLearning.AI / Andrew Ng (Coursera)
- **Mathematics for Machine Learning** – Imperial College London (Coursera)
- **OCI Generative AI Professional** – Oracle Cloud Infrastructure
- **OCI AI Foundations Associate** – Oracle Cloud Infrastructure
- **Professional Certificate** ×2 – Amity Future Academy
- **Short-term Courses** ×3 – NIELIT